

Exploring the Role of Contextual Belief in Sensory Integration and Motion Sickness for In-Car VR

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ABSTRACT

In in-car virtual reality (VR), ambiguous vehicle-related non-visual signals may increase reliance on top-down interpretation. We conducted a pilot study in a stationary vehicle using a headset-delivered low-frequency tone as a controllable surrogate cue, and manipulated instruction context to frame it as either aligned with VR forward motion (Aligned) or unrelated background (Not-Aligned). Phase 1 varied onset asynchrony to estimate temporal binding windows, and Phase 2 used a fixed asynchrony during repeated exposure to probe perceived coherence and motion sickness. This design offers an initial testbed for examining how contextual framing may shape multisensory inference and discomfort in in-car VR.

Index Terms: In-Car Virtual Reality; Priming; Motion sickness; Causal Inference; Autonomous vehicles

1 INTRODUCTION

In in-car virtual reality (VR), visually induced self-motion is primarily conveyed by the VR content, whereas non-visual vehicle signals such as engine sound or vibration originate from the car [4]. Motion sickness is often attributed to mismatches between visual motion and bodily cues [5], but in real driving these non-visual signals are inherently variable and frequently ambiguous. Under such uncertainty, perception relies more strongly on top-down beliefs and expectations when interpreting whether signals should be integrated [2].

We adopt a causal inference perspective [3] in which the brain estimates whether visual and vehicle-related cues share a common cause and should be integrated. In in-car VR, this inference is central because users must decide whether a given vehicle-related signal is meaningfully coupled to the virtual motion or merely coincidental noise. We therefore ask whether contextual beliefs about the vehicle–VR relationship can shift multisensory inference and, in turn, alter how perceived coherence relates to motion sickness.

To isolate belief-driven effects from physical variability, we conducted a pilot study in a stationary vehicle using a headset-delivered auditory surrogate cue, as an initial attempt to explore whether instructional framing about cue alignment with VR motion is reflected in temporal binding window (TBW) and in how perceived coherence relates to sickness under a fixed mismatch.

2 EXPERIMENT

2.1 Study design

We employed a 1×2 between-subject design to manipulate contextual framing about whether a vehicle-related cue should be interpreted as aligned with VR self-motion, to avoid carryover and mixing of frames within a session.

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Aligned: Participants were instructed to interpret the low-frequency tone delivered by the headset as a surrogate vehicle signal that corresponds to and is temporally aligned with the VR motion.

Not-Aligned: Participants were instructed to interpret the same tone as a surrogate vehicle cue that is not aligned with the VR motion and should be treated as an unrelated background.

The session consisted of two phases with complementary objectives. In Phase 1, participants performed a temporal-alignment judgment task in which we varied the stimulus onset asynchrony (SOA) between VR motion and the auditory surrogate delivered through the headset to estimate the temporal binding window (TBW) of each participant. In Phase 2, participants underwent repeated exposure under fixed asynchrony to induce motion sickness, while we repeatedly measured perceived coherence and sickness. This two-phase structure allowed us to characterize temporal tolerance separately from response to sickness.

To reinforce the manipulation, brief reminders were embedded throughout the session. In Phase 1, the reminder statements were presented as on-screen instructions between blocks. In Phase 2, the block-level survey items reiterated the assigned framing to maintain a stable interpretive context over time.

2.2 Experimental Set-up

The study was approved by the university's institutional review board (IRB) and six participants (2 female, 4 male; age: $M = 24.33$, $SD = 1.51$) participated in the study.

The experiment was conducted in a stationary vehicle. Participants wore a Meta Quest 3 and viewed a VR tunnel that induced self-motion via visual cues; no actual vestibular stimulation was provided. We delivered a controllable surrogate vehicle cue as headset-based low-frequency audio (100 Hz pure tone), with stimulus timing scheduled to control the onset asynchrony between visual motion and the tone across trials and phases.

2.3 Tasks and measures

Phase 1 estimated TBW while minimizing the carry-over to subsequent sickness induction. Between phases, the participants rested for at least 10 minutes. Before proceeding to Phase 2, we confirm that participants did not report motion sickness. After Phase 2, participants completed a framing manipulation check.

Phase 1: Temporal alignment and TBW. Phase 1 estimated tolerance to audiovisual timing offsets between VR-induced self-motion and the 100 Hz tone. The trials used seven SOAs (-500 , -300 , -150 , 0 , $+150$, $+300$, $+500$ ms; negative values indicate that the tone preceded motion onset). The visual motion burst followed a sinusoidal acceleration envelope with a peak of 10 m/s^2 and a period $T = 0.6 \text{ s}$ (Eq. 1). The tone amplitude followed the same envelope, and the SOA shifted only the onset timing. The participants judged whether the timing was consistent. The phase comprised three blocks, each presenting all SOAs five times in random order (105 trials total).

Phase 2: Sickness induction under fixed asynchrony. In Phase 2, participants repeatedly experienced visual motion with a fixed asynchrony between the VR motion and a 100 Hz tone to induce motion sickness under constant mismatch. The visual motion followed the same sinusoidal acceleration envelope with a peak

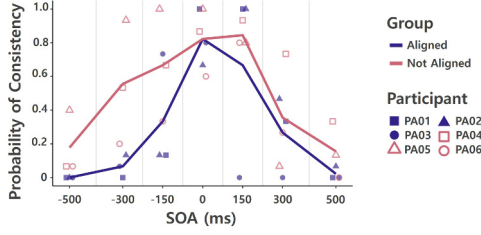


Figure 1: Phase 1 consistency judgments by SOA. Solid lines show group means, and markers show participant ratio for consistency across SOA levels.

of 10 m/s^2 but a longer period of 5 s to create sustained exposure (Eq. 1). The auditory cue was shifted by 300 ms and followed the same amplitude rule as in Phase 1, with amplitude proportional to the absolute acceleration magnitude (scaled to $[0, 1]$). The exposure was structured into five repeated motion cycles. After each cycle, participants completed a brief survey, generating five probe points in total during Phase 2, to report motion sickness.

$$a(t) = 10 \sin\left(\frac{2\pi t}{T_p}\right), \quad 0 \leq t < T_p, \quad T_p \in \{0.6, 5\} \text{ s}. \quad (1)$$

Measures. (1) *TBW*: full width at half maximum (FWHM) of a symmetric Gaussian fit to each participant's Phase 1 consistency-by-SOA curve. (2) *FMS*: Fast Motion Sickness questionnaire [1], rated on each Phase 2 probe on a scale of 0–20. (3) *Perceived coherence*: 7-point rating at each Phase 2 probe (1 = not integrated at all, 7 = very strongly integrated). (4) *Manipulation check*: post-Phase 2 7-point rating of perceived cue alignment with VR motion (1 = not aligned at all, 7 = highly aligned).

3 RESULTS

Given the pilot-scale sample ($N = 6$), we emphasize descriptive patterns and rely on summary from participants with exact permutation inference, avoiding strong distributional assumptions.

3.1 Belief Manipulation check.

For the manipulation check, ratings were higher in the Aligned group ($M = 5.33$, $SD = 0.58$) than in the Not-Aligned group ($M = 2.00$, $SD = 1.00$), suggesting that the instructed context was reflected in participants' interpretations.

3.2 Phase 1: Temporal alignment and TBW

Figure 1 shows Phase 1 consistency judgments across SOAs. For each participant, we fit a symmetric Gaussian to the consistency-by-SOA curve and computed TBW using the fitted function's FWHM. TBW was smaller on average in the Aligned group than in the Not-Aligned group (Aligned: $M = 294$ ms, $SD = 56$; Not-Aligned: $M = 572$ ms, $SD = 198$). With $N = 6$, an exact permutation test over all $\binom{6}{3} = 20$ label assignments yielded $p = 0.10$ (two-sided).

In general, Phase 1 provides preliminary evidence that *Aligned* framing is associated with a narrower TBW, suggesting a stricter temporal tolerance for offsets under identical stimulation.

3.3 Phase 2: Motion Sickness and Belief

As described in Figure 2, we examined whether Belief was related to sickness (FMS) on repeated probes. Repeated-measures correlation suggested a small association ($r = 0.23$, $p = 0.21$). Using slopes within-participants ($\text{FMS} \sim \text{Belief}$) to reduce variability between people, the mean slope differed in sign between groups (Aligned: -0.77 ; Not-Aligned: $+1.27$), with an exact permutation test over all $\binom{6}{3} = 20$ assignments yielding $p = 0.10$ (two-sided).

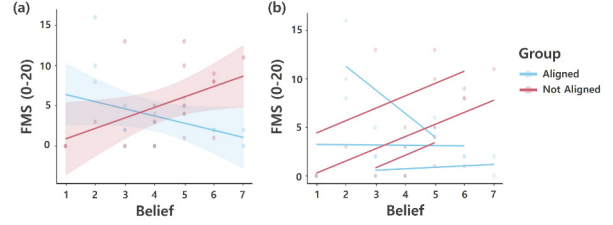


Figure 2: Phase 2 belief-sickness relationship. (a) Group-level linear fits ($\pm 95\%$ CI) of FMS as a function of belief. (b) Participant-level linear fits, colored by group.

This pattern was directionally consistent in leave-one-out checks (range $[-2.50, -1.21]$), suggesting a tentative context-dependent difference in how coherence relates to motion sickness.

In general, Phase 2 may suggest a context-dependent pattern under the same 300 ms mismatch: perceived coherence was associated with a lower sickness in the Aligned condition, while it was associated with a higher sickness in the Not-Aligned condition.

4 CONCLUSION

Taken together, Phases 1 and 2 suggest a coherent, belief-dependent pattern in multisensory inference under identical stimulation. In Phase 1, the Aligned context has the possibility that expecting coupled motion leads participants to apply stricter temporal constraints when judging whether the cues belong to the same event. In Phase 2, the coherence-sickness relationship differed by context under the same 300 ms mismatch. One interpretation is that, in the Aligned context, higher perceived coherence may reflect confirmation of the expected common cause and thus relate to lower sickness, whereas in the Not-Aligned context, high coherence may constitute an "unwanted binding" that violates the instructed interpretation and relates to higher sickness.

Given the small sample size, gender imbalance, and use of an auditory surrogate in a stationary vehicle, these results should be interpreted with caution. Future work with larger samples and richer vehicle-derived cues (e.g., vibration or motion) is needed for the robustness of these effects, to disentangle perceptual sensitivity from response criteria, and to clarify the mechanisms by which framing modulates multisensory integration and sickness in in-car VR.

ACKNOWLEDGMENTS

This work was supported by the MSIT (Korea) under the Global Research Support Program in the Digital Field (IITP; RS-2024-00431357), the IITP-ITRC program (IITP-2022-RS-2022-00156354), and the National Research Foundation of Korea (RS-2025-00514411).

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